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Scientific standards suffer from misrepresenting past work

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In "Our changing information ecosystem for science and why it matters for effective science communication," Krause et al. discuss the challenges of researching the current information environment (1). The lack of access to platform data is prominent among these challenges. We agree that data access is core to the research enterprise and that it is problematic to allow corporations to have so much power over data access (we have long argued similar things, e.g., refs. 2–4). However, we strongly disagree with the characterization of recent research (5–8) as "granting industry partners control over the coding of variables" (p. 5) or as being part of platform's investment "into long-term research streams that claim to contradict existing scholarship on phenomena like filter bubbles, in order to absolve their platforms of responsibility for adverse effects on society or individuals" (p. 7). These statements misrepresent the state of existing scholarship on filter bubbles and directly misrepresent the critiqued research. For instance, even a cursory read of ref. 5 reveals that its main conclusion is, in fact, the opposite of what Krause et al. imply. Among many other findings (5), reports that ideological segregation in news consumption on Facebook is three times as high as segregation on the web. How does such a finding, which inspired *Science*'s cover with a dark, ominous depiction of Meta under the headline "Wired to Split" (society), absolve the platform of responsibility?

We agree on the importance of "quality standards related to replicability and reproducibility" (p. 5). This is why studies (5–8) make all the code and data available for replication via the Inter-University Consortium for Political and Social Research, and advocate for data access to facilitate reproducibility, both within the articles and in related documentation (9). The authors also disclosed conflicts of interest throughout the review process and concurrently with publication (see ref. 9, frequently asked question #3). We are puzzled that Krause et al. do not critically engage the steps that (5–8) took to bolster research integrity, including the preregistration of research plans; the noninvolvement of Meta in the selection of external researchers; the coding and review of all data pipelines by these external researchers; and the control rights over the final version of the papers by non-Meta authors. These steps significantly improve on other known examples of industry collaboration in their ability to produce robust findings and far exceed the standards of most research typically published in academic journals, including many of the citations Krause et al. reference in ref. 1. We also agree that there is an essential power asymmetry between industry and researchers. We think it unlikely that any company will ever again voluntarily subject themselves to the independent scrutiny that Meta allowed during 2020. We are therefore very skeptical that companies will provide academics "gated" panels (as Krause et al. propose, p. 8). Instead, we believe that a combination of shared platform-independent data collection (10) and government-mandated data sharing (along the lines of the European Union's Digital Services Act), holds the most promise to address data access asymmetries.

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1. N. M. Krause, I. Freiling, D. A. Scheufele, Our changing information ecosystem for science and why it matters for effective science communication. *Proc. Natl. Acad. Sci. U.S.A.* **122**, e2400928121 (2025).
2. S. González-Bailón, *Decoding the Social World: Data Science and the Unintended Consequences of Communication* (MIT Press, Cambridge, MA, 2017).
3. D. Lazer *et al.*, Computational social science: Obstacles and opportunities. *Science* **369**, 1060–1062 (2020).
4. D. Lazer *et al.*, Meaningful measures of human society in the twenty-first century. *Nature* **595**, 189–196 (2021).
5. S. González-Bailón *et al.*, Asymmetric ideological segregation in exposure to political news on Facebook. *Science* **381**, 392–398 (2023).
6. A. M. Guess *et al.*, How do social media feed algorithms affect attitudes and behavior in an election campaign? *Science* **381**, 398–404 (2023).
7. A. M. Guess *et al.*, Reshares on social media amplify political news but do not detectably affect beliefs or opinions. *Science* **381**, 404–408 (2023).
8. B. Nyhan *et al.*, Like-minded sources on Facebook are prevalent but not polarizing. *Nature* **620**, 137–144 (2023), 10.1038/s41586-023-06297-w.
9. T. Stroud, J. A. Tucker, A. Franco, C. P. Kiewiet de Jonge, U.S. 2020 Facebook & Instagram Election Study (2023). https://medium.com/@2020_election_research_project/us-2020-facebook-instagram-election-study-frequently-asked-questions-faq-266d30cbe95b.
10. M. N. Meyer *et al.*, Enhancing the ethics of user-sourced online data collection and sharing. *Nat. Comput. Sci.* **3**, 660–664 (2023).